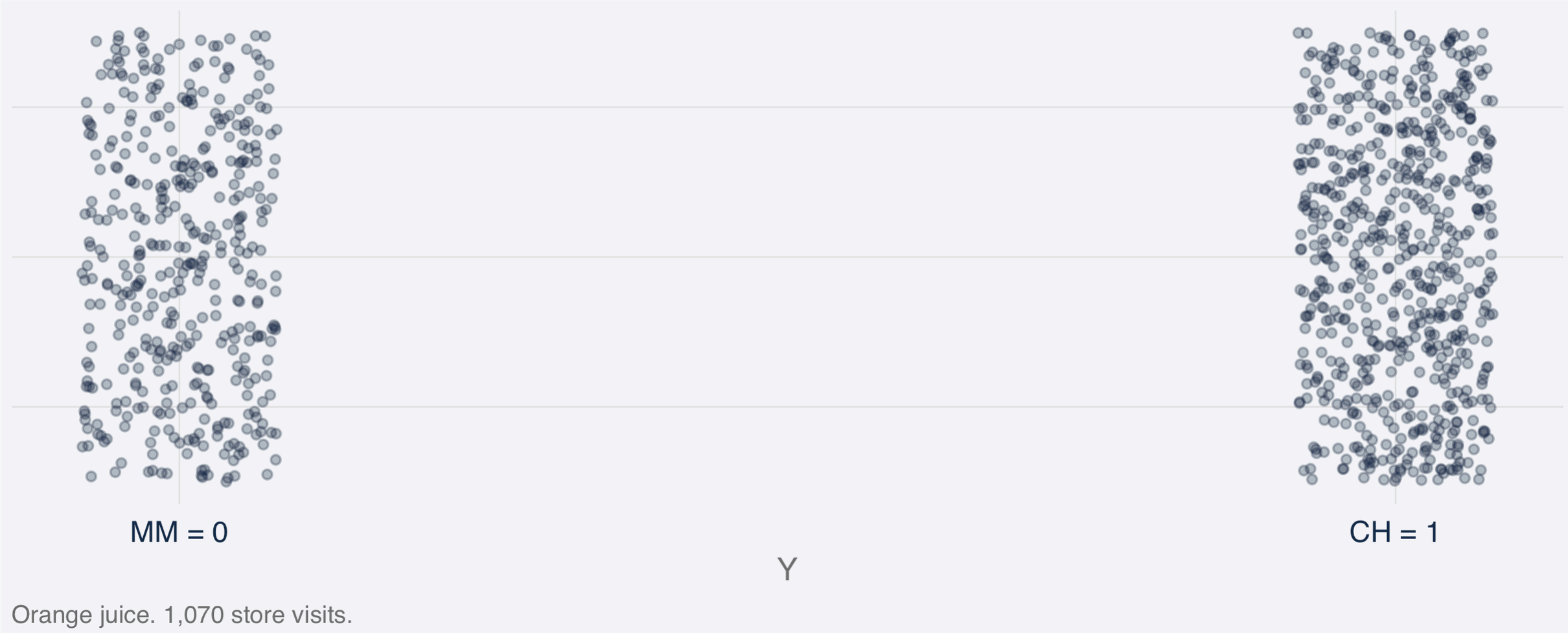
70-467 Machine Learning for Business Analytics

Logistic regression

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CMU-Q

Can we predict which juice a shopper buys?



Orange juice. 1,070 store visits.

The dataset has 1,070 store visits. Each one ends in Citrus Hill or Minute Maid.

Where are we on the course roadmap?

Topic	Description	Tools
PREDICT & EVALUATE		
Linear regression	Predict a number	OLS
Metrics	Size of the errors	RMSE, R^2
Train/test, leakage, cross-validation	Error on new data	Holdout, k-fold
Selection + regularization	Select variables or shrink coefficients	Best subset, ridge, lasso
CLASSIFY & DECIDE		
Logistic	**Probability of a brand**	**Logistic regression**
Thresholds	Turn a probability into a call	ROC, cutoff
FLEXIBLE & UNSUPERVISED		
Trees	Split the predictors in sequence	Decision trees
Forests / boosting	Combine many trees	Random forest, boosting
PCA	Fewer combinations of the predictors	Principal components
Clustering	Group similar observations	K-means

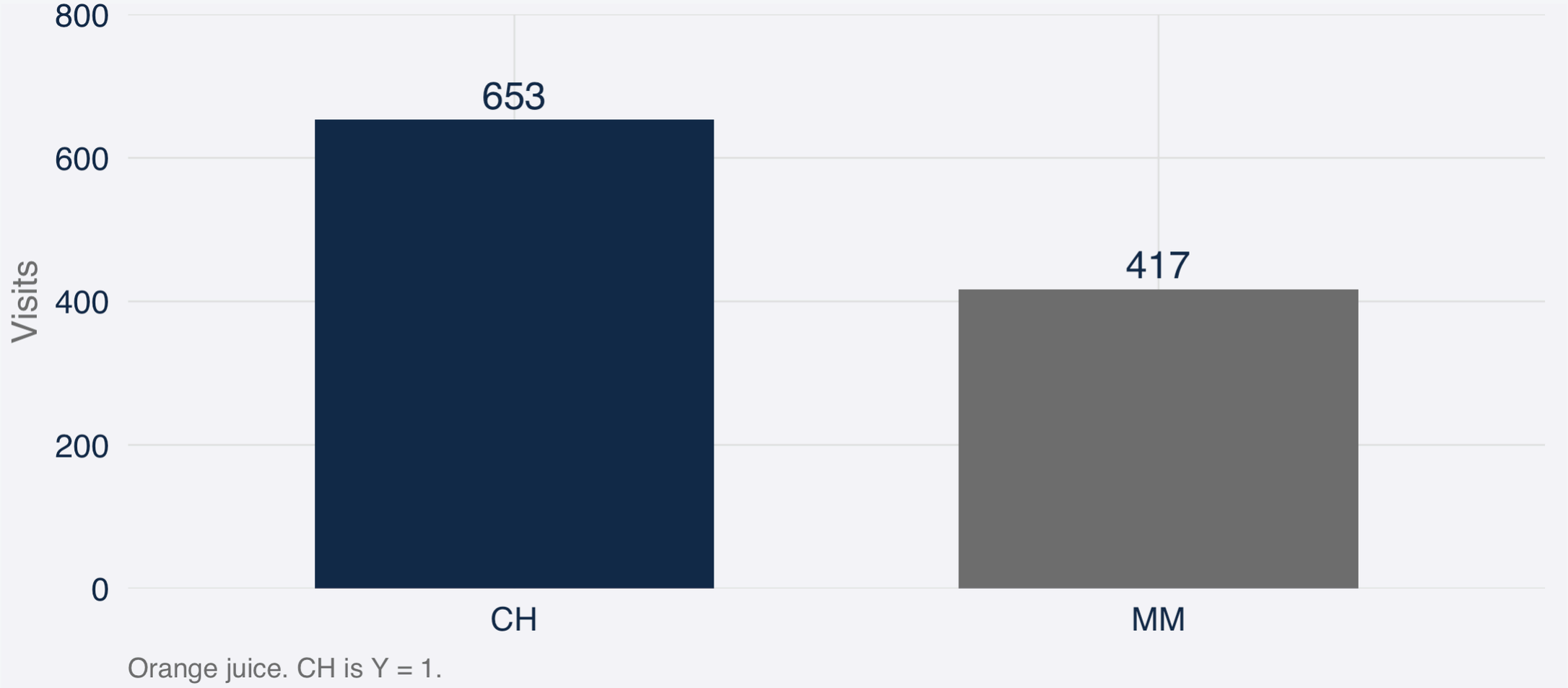
Before we have dealt with problems where the outcome was a continuous variable, like predicting a credit card balance, the price of a house, etc, and how do we decide which predictors to keep.

This week the outcome is the brand on the visit, Citrus Hill or Minute Maid. We want the probability the visit bought Citrus Hill.

What we will do

1. The linear probability model: least squares of a 0/1 outcome
2. Logistic regression: a model for the p(Y = 1 | X)
3. Making sense of the coefficients in logistic regression: the log-odds, the odds, and the changes in the probabilities
4. Estimating the probabilities for a given visit
5. Assessing the Classification accuracy **— make-up**

How is the brand coded?



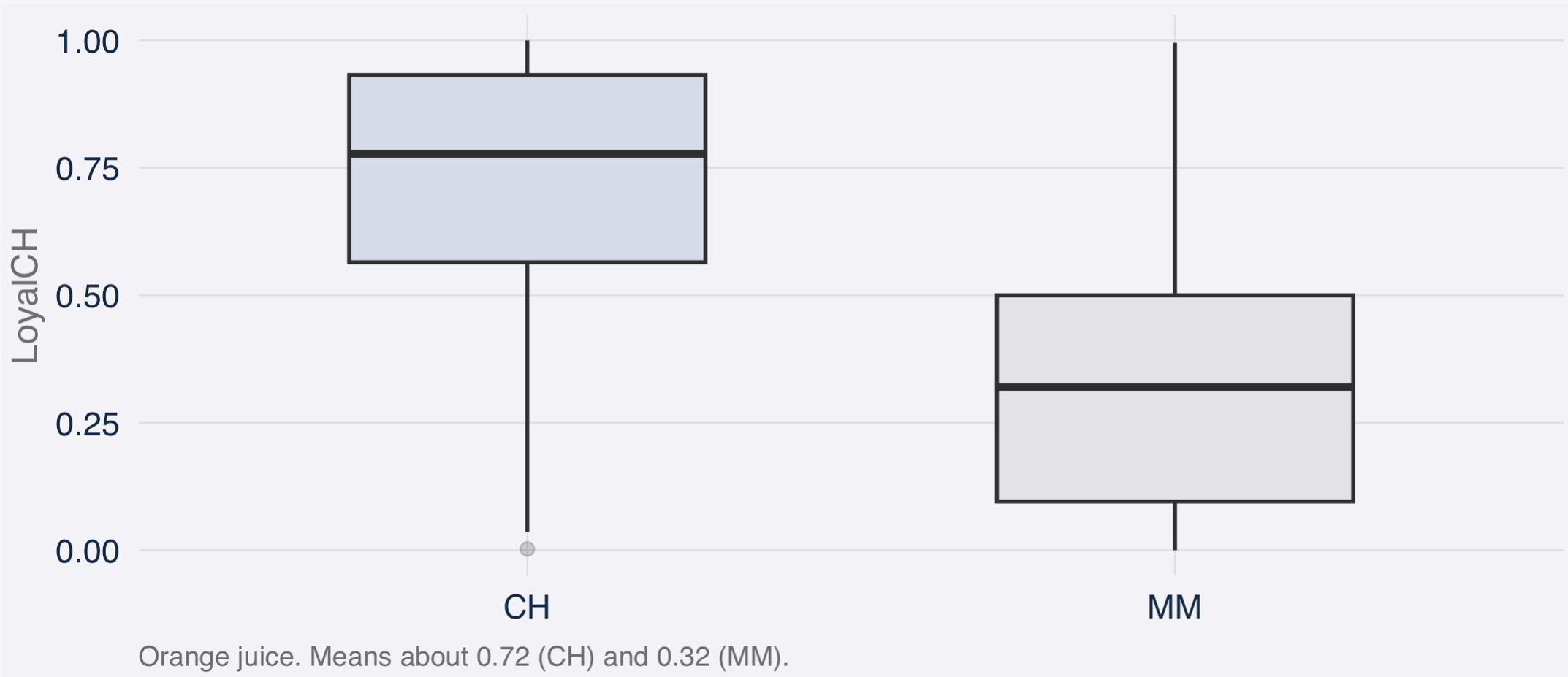
653 of 1,070 visits bought CH, 61 percent. The other 417 bought MM. CH is Y = 1.

Codebook for the dataset?

Variable	Type	Unit	Description
Purchase	text	CH or MM	Brand bought on the visit. This is th outcome. We code Y = 1 for CH and = 0 for MM.
LoyalCH	numeric	score, 0 to 1	Loyalty to CH, already in the dataset Used in the prediction.
SalePriceMM	numeric	dollars	Shelf price of MM. Used only to buil PriceDiff.
SalePriceCH	numeric	dollars	Shelf price of CH. Used only to build PriceDiff.
PriceDiff	numeric	dollars	SalePriceMM minus SalePriceCH. Positive when MM costs more. Used in the prediction.

Two variables enter the prediction: LoyalCH, and PriceDiff, which is the MM price minus the CH price.

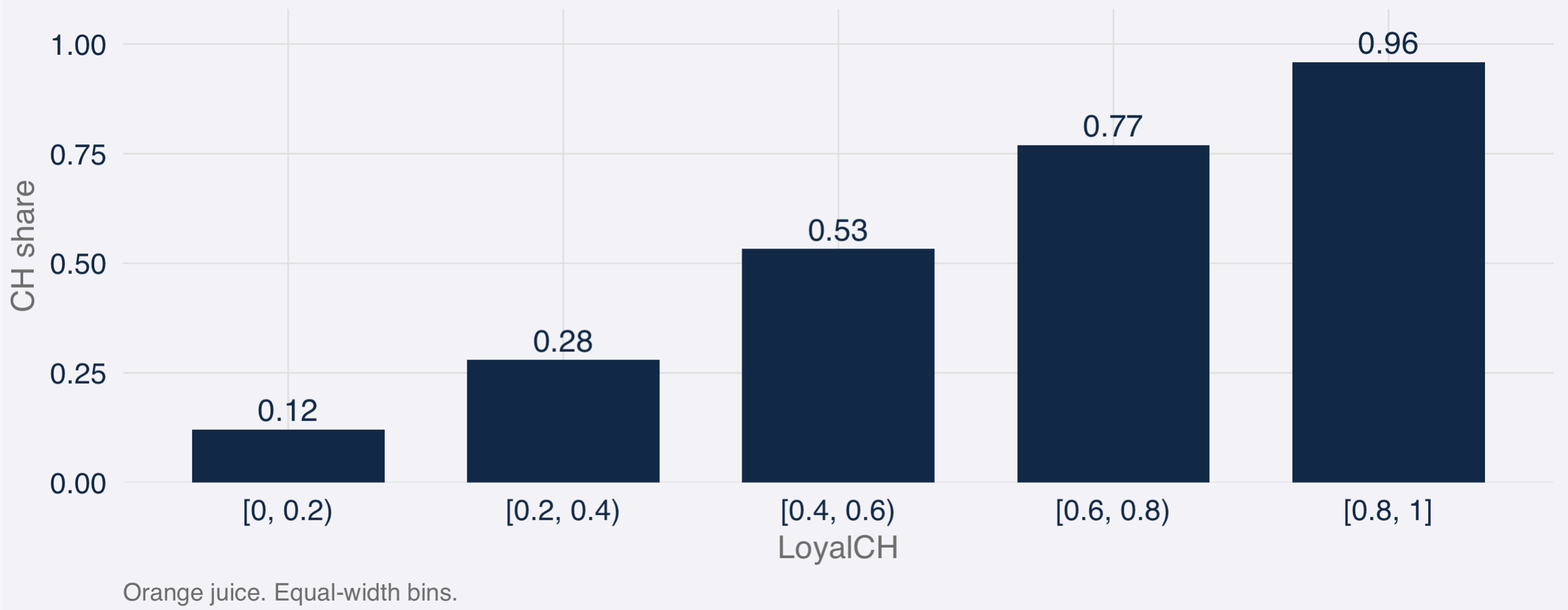
Does loyalty separate the two brands?



Orange juice. Means about 0.72 (CH) and 0.32 (MM).

LoyalCH averages about 0.72 on CH visits and about 0.32 on MM visits.

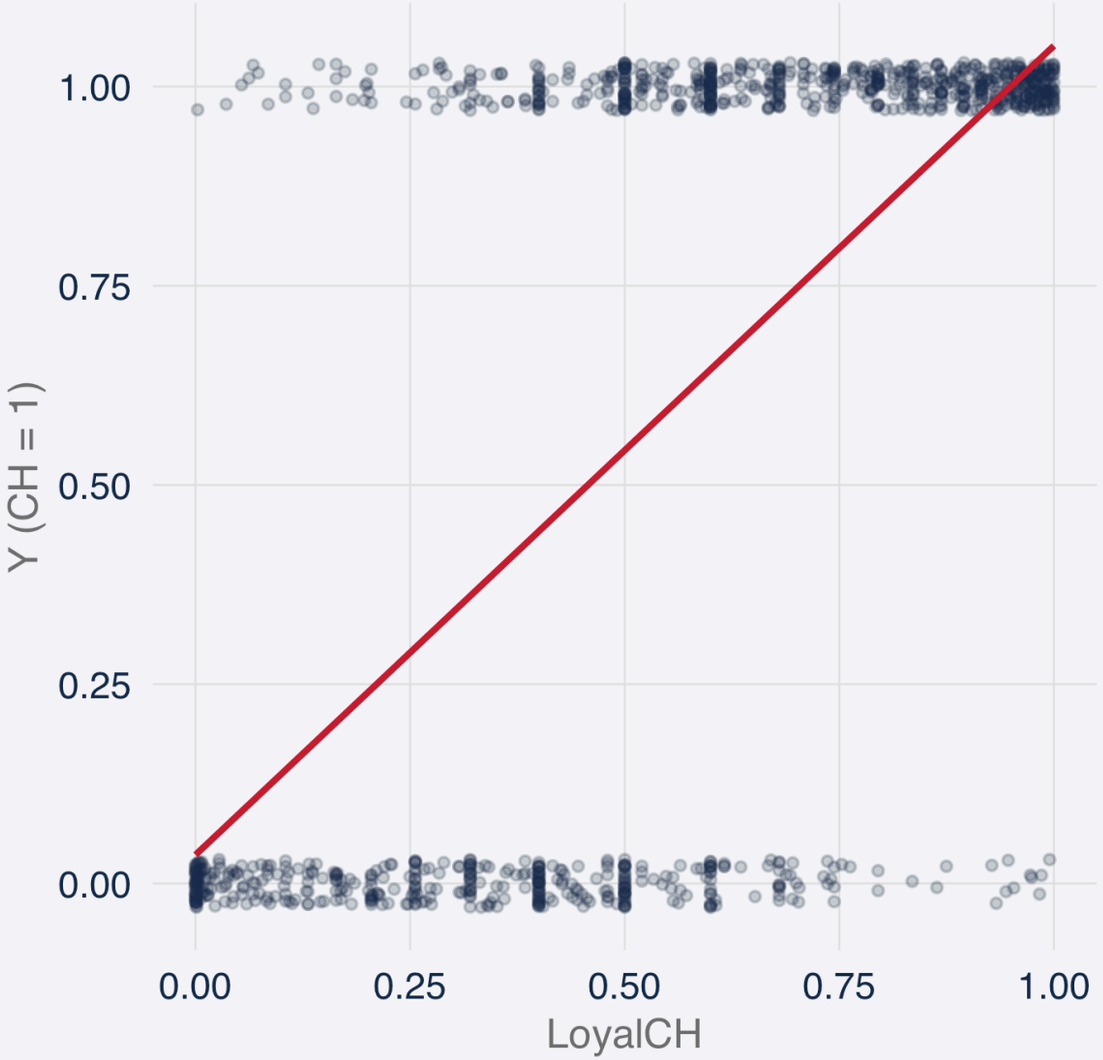
How does the CH share change across loyalty bins?



Orange juice. Equal-width bins.

In the five bins, the share of visits that bought CH is about 0.12, 0.28, 0.53, 0.77, and 0.96.

The linear probability model



Orange juice. Least squares of Y on LoyalCH.

- Least squares of Y on LoyalCH:

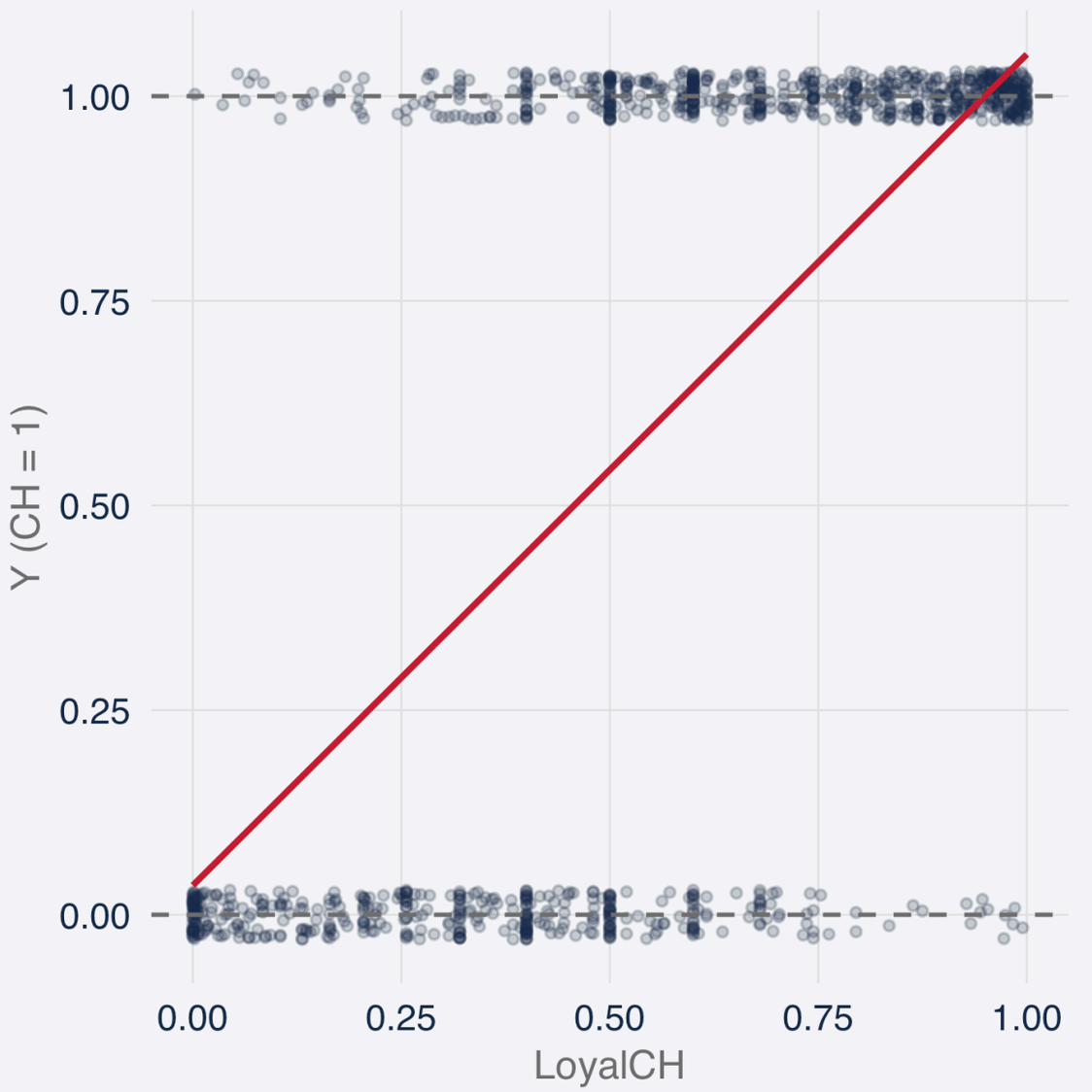
$$p(Y = 1 \mid \text{LoyalCH}) = b_0 + b_1 \, \text{LoyalCH}$$

  b_1 is the change in the probability of CH when LoyalCH goes from 0 to 1.
- Suppose $b_1 = 0.8$. A customer whose LoyalCH is 0.1 higher has a probability buying CH that is 0.08 higher, or 8 percentage points.
- For measuring effects, economists still use it. Because

$$E(Y \mid X) = p(Y = 1 \mid X),$$

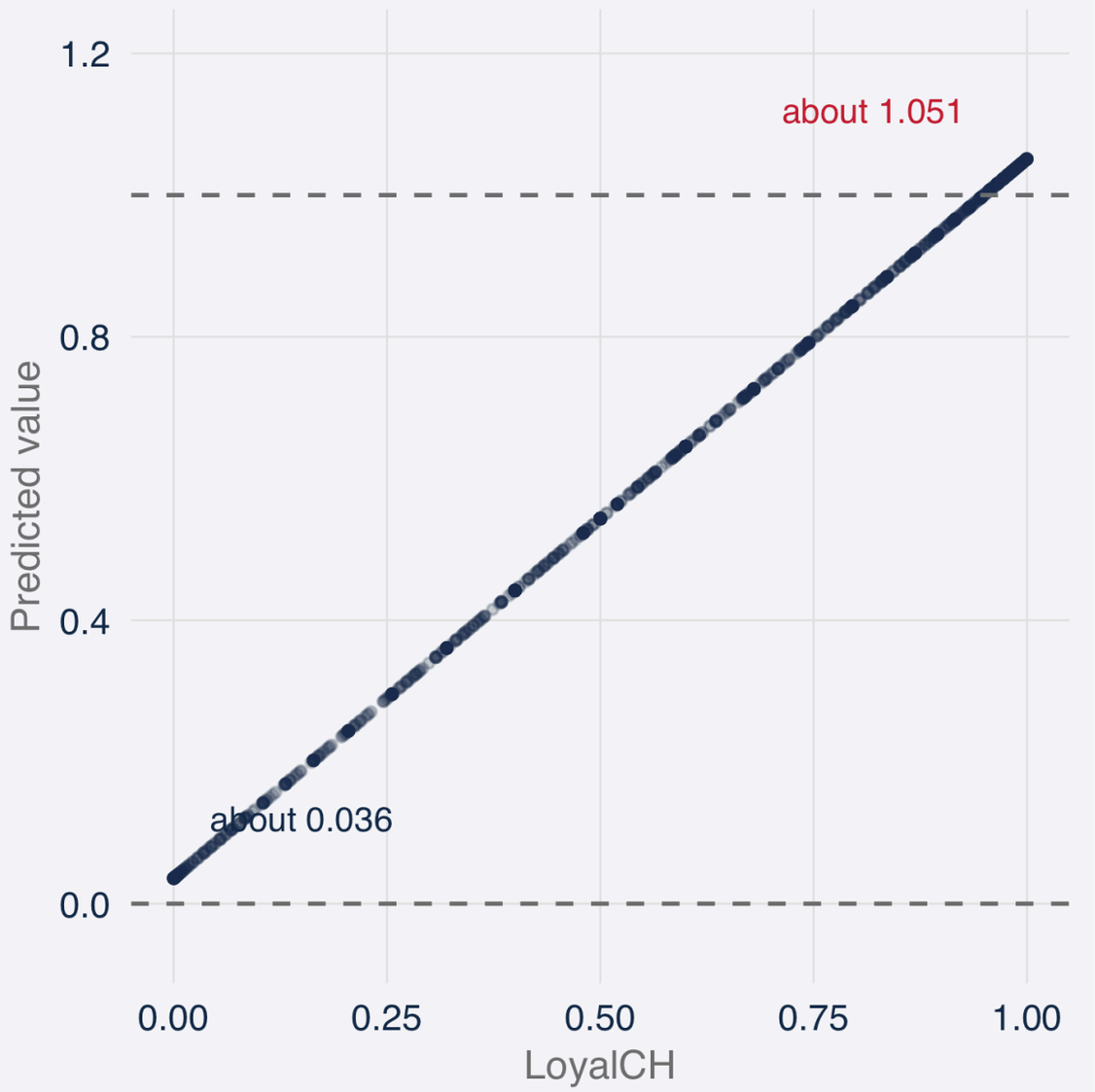
  b_1 is the effect of X on the probability. In a randomized job-training program, $b_1 = 0.05$ means training raises the probability of employment by 5 percentage points.
- For prediction, it is weak. Nothing keeps $b_0 + b_1 X$ between 0 and 1, so a predicted probability can be below 0 or above 1.

What is wrong with the linear probability model?



- **The errors are heteroskedastic.** The error variance is $p(1-p)$, so it chang with LoyalCH. The usual standard errors are wrong; robust standard errors needed.
- **The errors are not normal.** Y is only 0 or 1, so at a given LoyalCH the error takes just two values, $1-p$ or $-p$.
- **A predicted value can fall outside 0 and 1.** Nothing bounds the line. The dashed lines mark 0 and 1, and the line crosses 1 at high LoyalCH.

How far outside 0 and 1 does the linear probability model go?



- The predicted values run from about 0.036 to 1.051. The line is

$$0.036 + 1.015 \times \text{LoyalCH}.$$

- At loyalty 0 the calculation is

$$0.036 + 1.015 \times 0 = 0.036.$$

  That value is still inside 0 and 1.

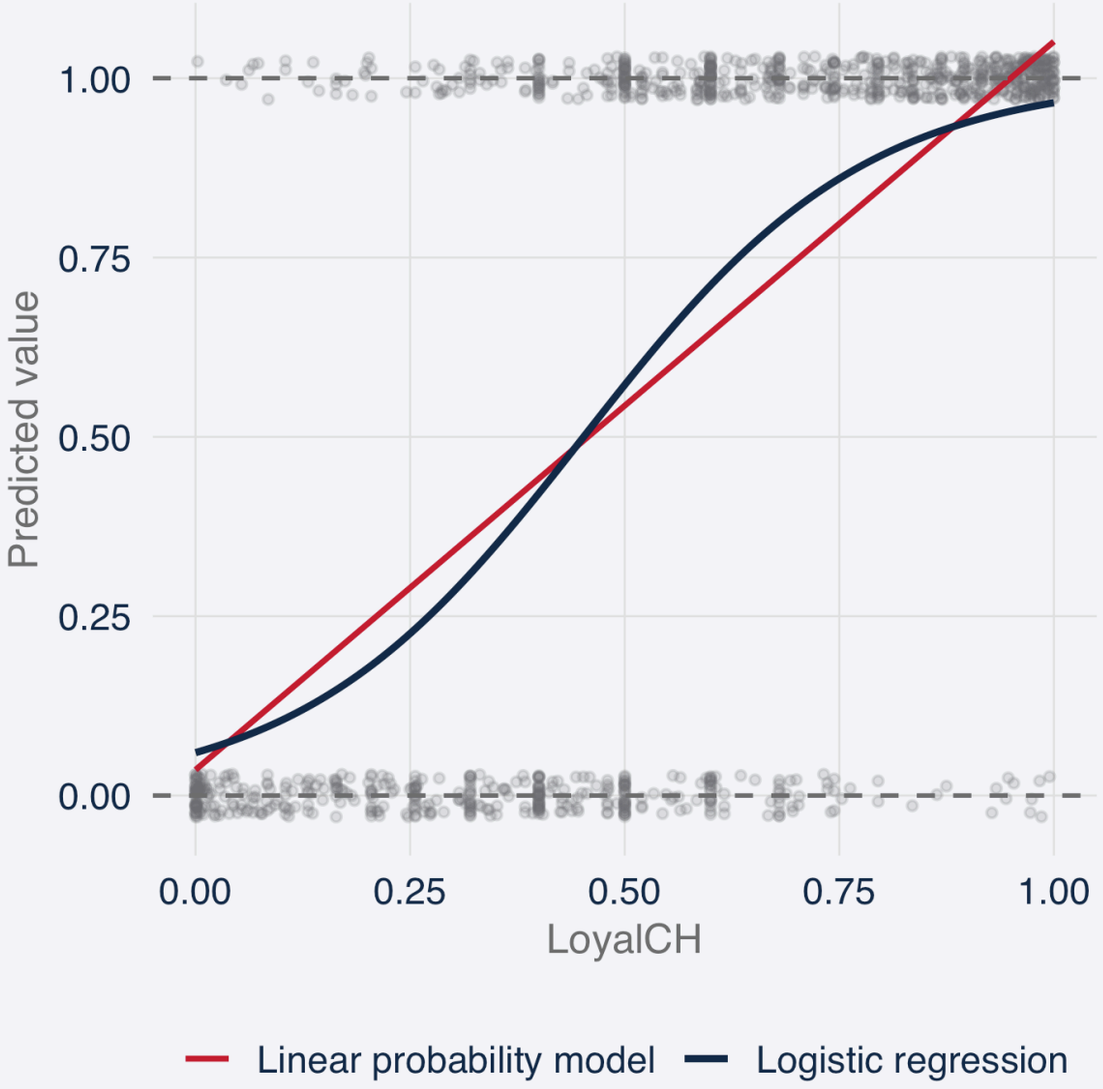
- At loyalty 1 the calculation is

$$0.036 + 1.015 \times 1 = 1.051.$$

  That value is above 1, so it is not a probability of CH.

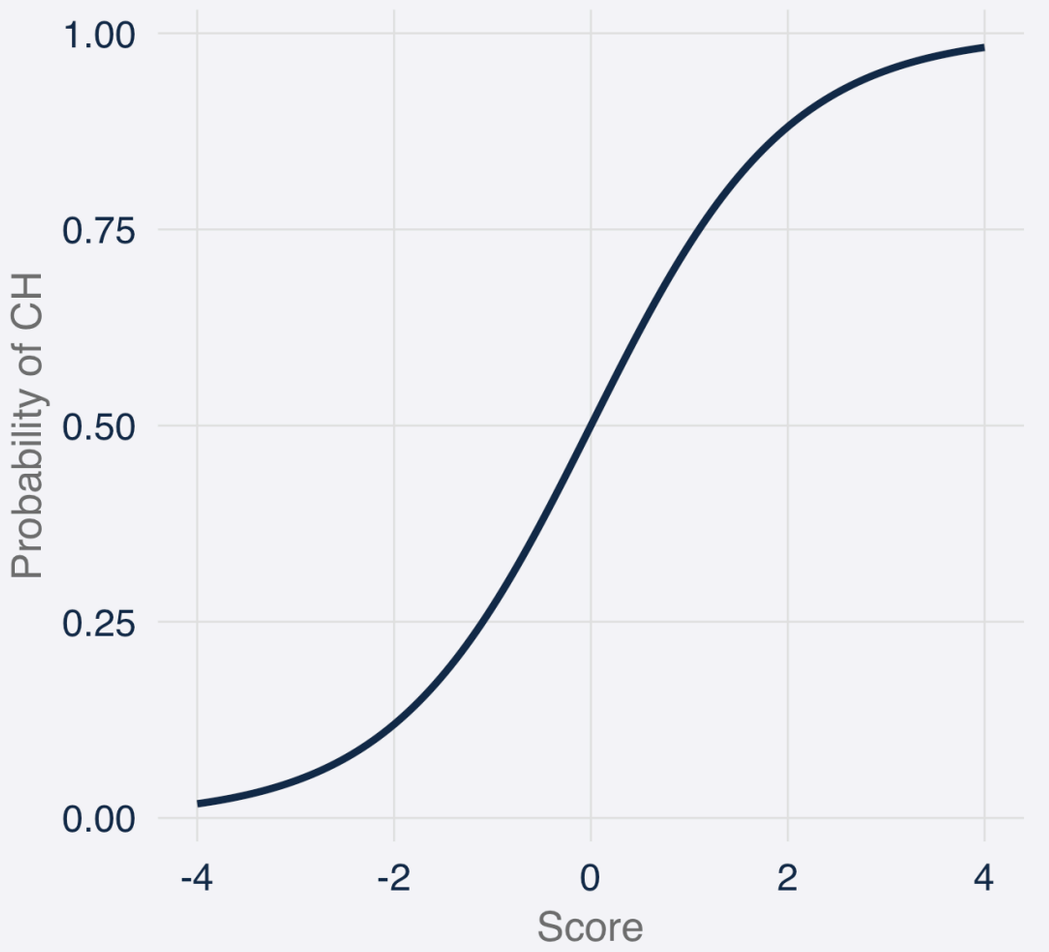
- Suppose the store uses these values to decide which shoppers get a Minute Maid coupon. Why could a predicted value of 1.051 be a problem?

The linear probability model and logistic regression



- The red line is least squares of Y on LoyalCH. Above 1, that fitted value is not probability.
- The navy curve is logistic regression on the same loyalty. Every point on it is a probability of CH between 0 and 1.

Logistic regression puts a line on the log-odds of CH



- The score is this line in LoyalCH. It is not yet a probability of CH:

$$\text{score} = b_0 + b_1 \times \text{LoyalCH}.$$

- We do not fit Y. The curve turns the score into the probability of CH:

$$p = \frac{e^{\text{score}}}{1 + e^{\text{score}}}.$$

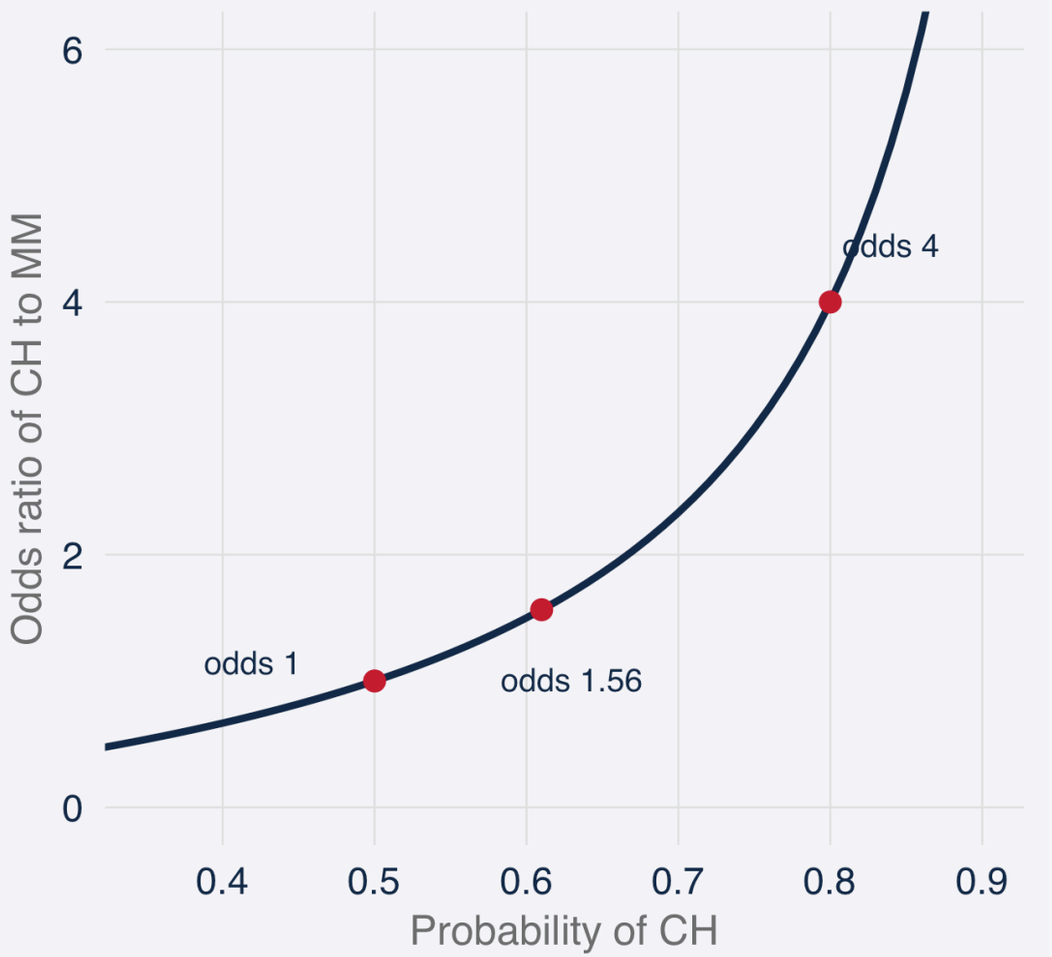
  It stays between 0 and 1.
- The odds are the probability divided by one minus the probability. On this curve those odds equal e to the score:

$$\frac{p}{1-p} = e^{\text{score}}.$$

- Take the log of both sides. The score is that log-odds. With loyalty alone the LoyalCH coefficient is 6.09, so LoyalCH higher by 1 raises the log-odds of CH by 6.09:

$$\log\left(\frac{p}{1-p}\right) = b_0 + 6.09 \times \text{LoyalCH}.$$

What is the odds ratio of CH to MM?



- The odds on the last slide are the odds ratio of CH to MM. That ratio is the probability of CH divided by the probability of MM:

$$\frac{p}{1-p}.$$

- To go from the odds ratio back to the probability, divide the odds ratio by 1 plus t odds ratio:

$$p = \frac{\text{odds ratio}}{1 + \text{odds ratio}}.$$

- An odds ratio of 1 gives a probability of 1/2. An odds ratio of 4 gives a probability 0.8:

$$\frac{1}{1+1} = \frac{1}{2}, \qquad \frac{4}{1+4} = 0.8.$$

- In this file the CH share is 0.61. The odds ratio is about 1.56, and that odds ratio comes back to a probability of about 0.61:

$$\frac{0.61}{0.39} \approx 1.56, \qquad \frac{1.56}{1+1.56} \approx 0.61.$$

How does logistic regression give the probability of CH from loyalty?

$$p(Y = 1 \mid \text{LoyalCH}) = \frac{e^{b_0 + b_1 \, \text{LoyalCH}}}{1 + e^{b_0 + b_1 \, \text{LoyalCH}}}$$

- `glm(Y ~ LoyalCH, family = binomial)` gives the estimated score:

$$\text{score} = -2.76 + 6.09 \times \text{LoyalCH}.$$

- Put the score into the formula above. At loyalty 0 the score is −2.76:

$$p = \frac{e^{-2.76}}{1 + e^{-2.76}} \approx 0.06.$$

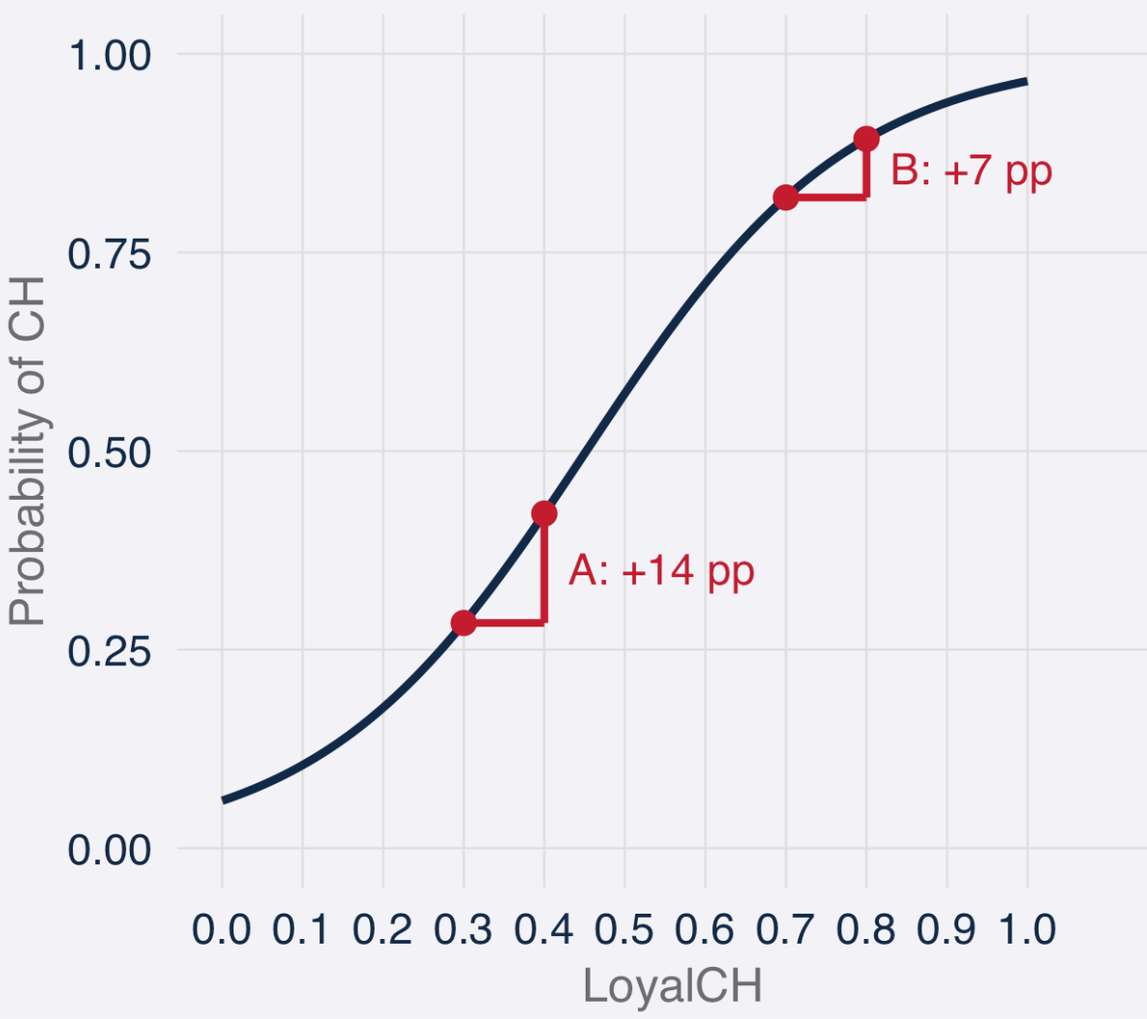
- **The intercept.** A shopper with LoyalCH of 0 buys CH with a probability of about 0.06.
- **The coefficient in log-odds.** 6.09 is positive, so higher loyalty means a higher probability of CH. LoyalCH higher by 0.1 always raises the log-odds of CH by 6.09 × 0.1 ≈ 0.61.
- **The coefficient in probability.** The same 0.1 does not add a fixed amount to the probability. The next slide shows why.

What does 0.1 more loyalty do to the probability of CH?

$$\text{score} = -2.76 + 6.09 \times \text{LoyalCH} \qquad p = \frac{e^{\text{score}}}{1 + e^{\text{score}}}$$

Step	LoyalCH	Score	Probability of CH
A	0.3 → 0.4	−0.93 → −0.32	0.28 → 0.42
B	0.7 → 0.8	1.50 → 2.11	0.82 → 0.89

- The score rises by 6.09 × 0.1 ≈ 0.61 in both steps.
- The probability rises by 14 percentage points in step A (28% to 42%), but only 7 percentage points in step B (82% to 89%).
- In log-odds, 0.1 more loyalty always adds 0.61. In probability, the change depends on the shopper's starting loyalty.



What does 0.1 more loyalty do to the odds?

$$\text{odds} = \frac{p}{1-p} = e^{\text{score}} \qquad e^{0.61} \approx 1.84$$

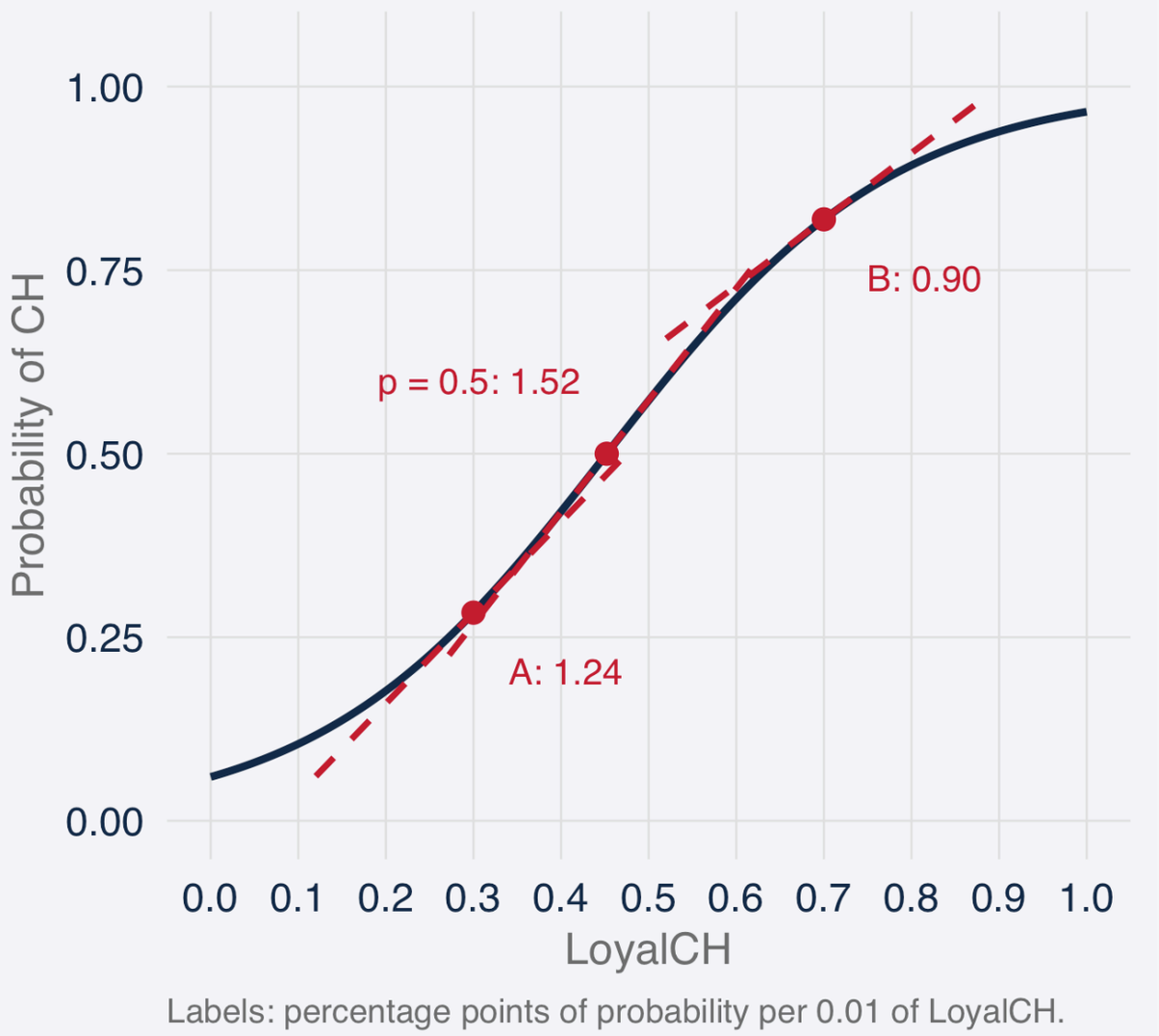
Step	LoyalCH	Score	Odds of CH	Probability of CH
A	0.3 → 0.4	−0.93 → −0.32	0.39 → 0.73	0.28 → 0.42
B	0.7 → 0.8	1.50 → 2.11	4.48 → 8.25	0.82 → 0.89

- Both steps add 0.61 to the score, so both multiply the odds of CH by about 1.84, as in the box.
- A step of any size Δ in LoyalCH multiplies the odds by

$$e^{6.09 \times \Delta}.$$

- The odds multiplier is the same for every shopper. The change in the probability is not.

Why does step A move the probability more than step B?



Labels: percentage points of probability per 0.01 of LoyalCH.

- The slope of the curve at a point with probability p is

$$6.09 \times p \times (1 - p).$$

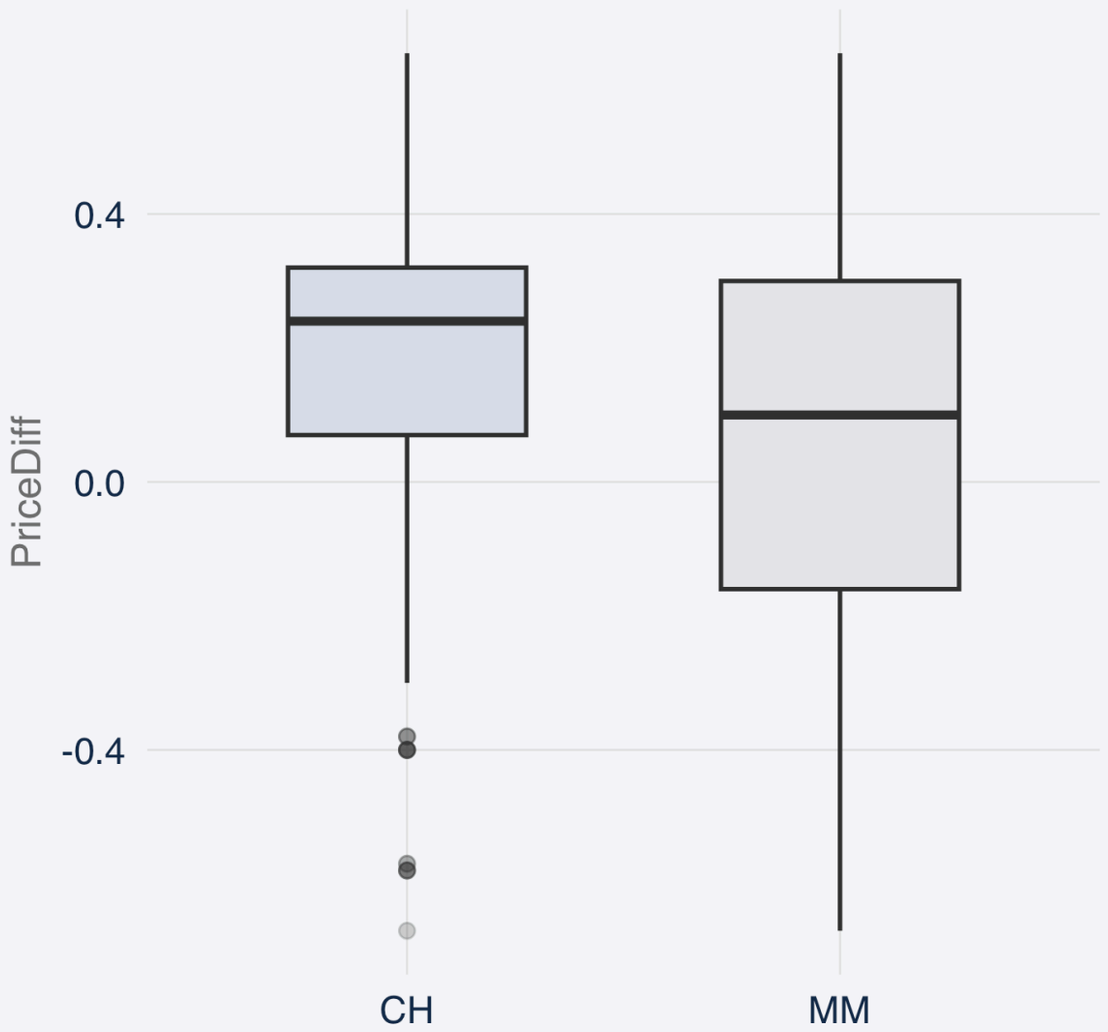
  It is the change in the probability per unit of LoyalCH. Divide by 100 to read it per 0.01 of loyalty.
- At step A, p = 0.28 and the slope is about 1.24. Each extra 0.01 of loyalty raises the probability of CH by about 1.24 percentage points.
- At p = 0.5 the slope is largest, about 1.52, or 1.52 points per 0.01.
- At step B, p = 0.82 and the slope is about 0.90, or 0.90 points per 0.01. Th same loyalty moves this shopper less.

How do we read the LoyalCH coefficient of 6.09?

Reading	Rule for a step Δ in LoyalCH	For 0.1 more loyalty	Same for every shopper?
Log-odds	add 6.09 × Δ	+0.61	Yes
Odds	multiply by $e^{6.09 \times \Delta}$	×1.84	Yes
Probability	compute p before and after	A: 0.28 → 0.42, +14 pp B: 0.82 → 0.89, +7 pp	No

- Log-odds and odds give one number for everyone, but nobody thinks in log-odds or odds.
- The probability is what a manager needs, but it depends on where the shopper starts. Its slope, 6.09 × p × (1 − p), is called the **marginal effect** of LoyalCH.
- How to report it: “For a shopper at LoyalCH 0.3, 0.1 more loyalty raises the probability of CH by about 14 percentage points.”

Does the price gap differ between CH visits and MM visits?



PriceDiff = SalePriceMM – SalePriceCH. Means about 0.21 (CH) and 0.05 (MM).

- PriceDiff is the MM price minus the CH price.
- It averages about 0.21 on CH visits and about 0.05 on MM visits.

How do we fit both variables in R?

```r
fit2 <- glm(Y ~ LoyalCH + PriceDiff, data = oj, family = binomial)   # logistic regression: Y = 1 if CH
round(coef(fit2), 2)                                                 # intercept and the two slopes
round(exp(coef(fit2)[-1] * 0.1), 2)                                  # odds multiplier for +0.10 loyalty and +$0.10 gap
```

```
(Intercept)    LoyalCH   PriceDiff
      -3.25       6.40       2.86
   LoyalCH PriceDiff
      1.90      1.33
```

- **Score:** $\text{score} = -3.25 + 6.40 \times \text{LoyalCH} + 2.86 \times \text{PriceDiff}$.
- **Intercept,** -3.25**.** At $\text{LoyalCH} = 0$ and $\text{PriceDiff} = \$0.00$, the log-odds of CH are -3.25, the odds are $e^{-3.25} \approx 0.039$, an the probability of CH is $\frac{0.039}{1+0.039} \approx 0.04$ (4%).
- **LoyalCH,** $6.40 > 0$ **(0 to 1 scale).** Holding PriceDiff fixed, $+0.10$ in LoyalCH adds $6.40 \times 0.10 = 0.64$ to the log-odds of CH, multiplies the odds of CH by $e^{0.64} \approx 1.90$ ($+90\%$), and **increases** $P(Y = 1)$.
- **PriceDiff,** $2.86 > 0$ **(dollars:** $\text{SalePriceMM} - \text{SalePriceCH}$**).** Holding LoyalCH fixed, a $+\$0.10$ (10-cent) wider price gap ad $2.86 \times 0.10 \approx 0.29$ to the log-odds of CH, multiplies the odds of CH by $e^{0.286} \approx 1.33$ ($+33\%$), and **increases** $P(Y = 1)$.

What is the marginal effect for one shopper?

$$\text{marginal effect of } x_j = b_j \times p \times (1 - p)$$

```r
shopper <- data.frame(LoyalCH = 0.5, PriceDiff = 0)        # the shopper we pick: medium loyalty, equal prices
p0 <- unname(predict(fit2, shopper, type = "response"))   # P(CH) for this shopper
round(c(p = p0,                                            # print p ...
        coef(fit2)[-1] * p0 * (1 - p0)), 2)                # ... then b x p x (1 - p) for each variable
```

```
    p  LoyalCH PriceDiff
 0.49     1.60     0.72
```

Variable	b_j	$p(1-p)$	Marginal effect (per 1 unit)	Per realistic step
LoyalCH	6.40	0.49 × 0.51 ≈ 0.25	6.40 × 0.25 ≈ 1.60	+0.10 loyalty: **+16 pp**
PriceDiff	2.86	0.25	2.86 × 0.25 ≈ 0.72	+$0.10 gap: **+7.2 pp**

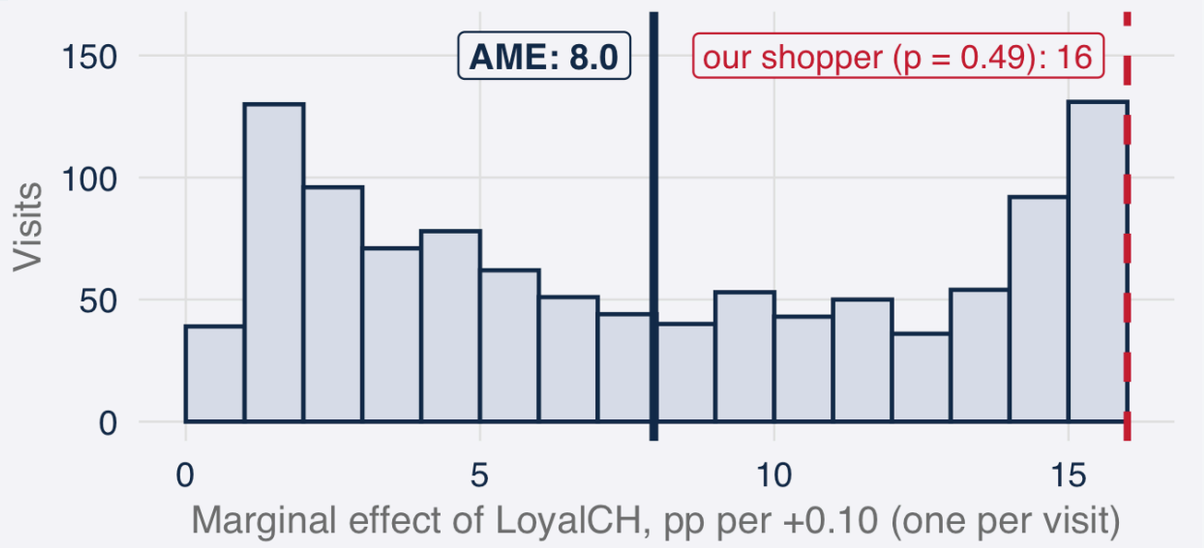
- **Our shopper:** LoyalCH 0.5, equal prices, $p = 0.49$.
- **LoyalCH:** 0.10 more loyalty raises her probability of CH by about **16 percentage points**.
- **PriceDiff:** MM costing $0.10 more raises it by about **7.2 percentage points**.

What is the average marginal effect (AME) across all shoppers?

$$\text{AME of } x_j = \frac{1}{n}\sum_{i=1}^{n} b_j \times p_i \times (1 - p_i) = \text{average of the } n = 1{,}070 \text{ marginal effects, each at its own } p_i$$

```r
p        <- predict(fit2, type = "response")        # P(CH) for each of the 1,070 visits
me_loyal <- coef(fit2)["LoyalCH"]   * p * (1 - p)   # marginal effect of LoyalCH, one per visit
me_gap   <- coef(fit2)["PriceDiff"] * p * (1 - p)   # marginal effect of PriceDiff, one per visit
round(c(LoyalCH = mean(me_loyal), PriceDiff = mean(me_gap)), 3)   # AME = average of each
```

```
LoyalCH PriceDiff
  0.795     0.356
```



- **LoyalCH:** 0.795 × 0.10 ≈ **+8.0 pp** per +0.10 loyalty.
- **PriceDiff:** 0.356 × 0.10 ≈ **+3.6 pp** per +$0.10 gap.
- **Why 8.0, not last slide's 16:** our shopper was undecided, close to 50/50. Many shoppers are almost sure to buy CH or MM, so more loyalty barely moves them. That lowers the average.

> **Your turn:** when would you report the marginal effect for one shopper, and when the AME?

When do we use the marginal effect and when the AME?

It depends on who the decision is for. Take the Citrus Hill brand manager with a 10-cent price cut, which raises PriceDiff by $0.10

Marginal effect: one customer or segment

- **Question:** who should get a 10-cent CH coupon?
- **Undecided shopper** (LoyalCH 0.5, $p = 0.49$): **+7.2 pp**.
- **Loyal shopper** (LoyalCH 0.9, $p = 0.92$): **+2.0 pp**. She buys CH anyway.
- **Decision:** send coupons to undecided shoppers, not loyal ones.

AME: a policy for everyone

- **Question:** what does a 10-cent CH price cut in every store do?
- **Every visit** gets the cut, so average over all 1,070: **+3.6 pp**.
- **In sales:** $0.036 \times 1{,}070 \approx 38$ more CH purchases per 1,070 visits.
- **Decision:** compare those 38 extra sales with the cost of 10 cents on every CH sale.

Deciding for one customer or segment: marginal effect at their p. Deciding for everyone at once: the AME.